Here, random variable X denotes the number on the first ticket and random variable Y denotes the number on the second ticket.

9.5.2 Joint and marginal probability density functions

As we defined in section 9.5.1 the joint probability mass function, we define the joint probability density function.

Definition:

Let (X, Y) be a bivariate continuous random variables. The function $f(x, y)$ is called a bivariate probability density if the following conditions are satisfied.

(i) $f(x, y) \geq 0 \qquad \forall\ x, y$

(ii) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)dx\,dy = 1$

The marginal probability density function of X is given by

$$g(x) = \int_{-\infty}^{\infty} f(x, y)dy$$

and the marginal probability density function of Y is given by

$$h(y) = \int_{-\infty}^{\infty} f(x, y)dx$$

Example 9.7

Prove that the bivariate function given by $f(x, y) = \begin{cases} \frac{1}{8}(x+y) & , \quad 0 < x, y \leq 2 \\ 0 & , \quad otherwise \end{cases}$

Proof:

If f is a probability density function $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)dx\,dy = 1$

$$= \frac{1}{8}\int_{0}^{2}\int_{0}^{2}(x+y)dx\,dy$$

$$= \frac{1}{8}\left(\int_{0}^{2}\int_{0}^{2} xdx\,dy + \int_{0}^{2}\int_{0}^{2} ydx\,dy\right)$$

$$= \frac{1}{8}\left\{\int_{0}^{2}\left[\int_{0}^{2} xdx\right]dy + \int_{0}^{2}\left[\int_{0}^{2} ydy\right]dx\right\}$$

$$= \frac{1}{8}\left[\int_{0}^{2} 2dy + \int_{0}^{2} 2dx\right]$$